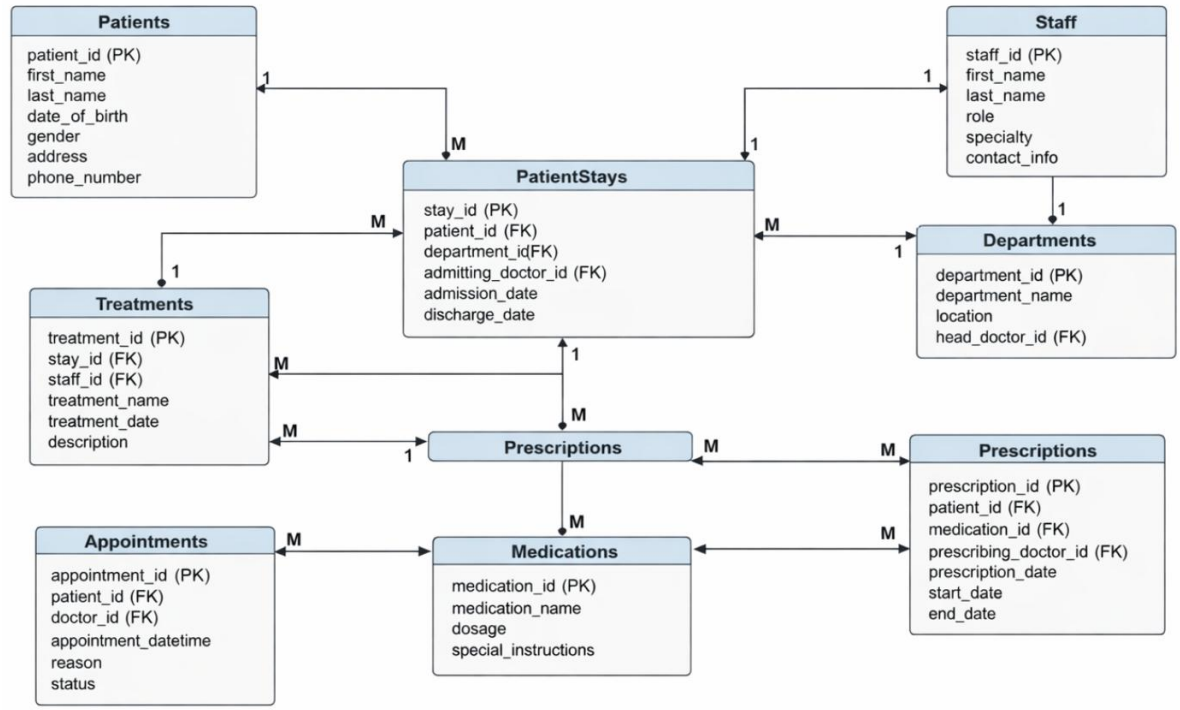
Er-kaavio


Patients
patient_id (PK)
first_name
last_name
date_of_birth
gender
address
phone_number
Staff
staff_id (PK)
first_name
last_name
role
specialty
contact_info
PatientStays
stay_id (PK)
patient_id (FK)
department_id(FK)
admitting_doctor_id (FK)
admission_date
discharge_date
Departments
department_id (PK)
department_name
location
head_doctor_id (FK)
Treatments
treatment_id (PK)
stay_id (FK)
staff_id (FK)
treatment_name
treatment_date
description
Prescriptions
Prescriptions
prescription_id (PK)
patient_id (FK)
medication_id (FK)
prescribing_doctor_id (FK)
prescription_date
start_date
end_date
Appointments
appointment_id (PK)
patient_id (FK)
doctor_id (FK)
appointment_datetime
reason
status
Medications
medication_id (PK)
medication_name
dosage
special_instructions
1
M

Tiedon syöttö

```
"C:\UniServerZ\core\mysql\bin\mysql.exe" -u root [redacted] hospital -e "SELECT * FROM Patients;"
+------------+------------+------------+---------[redacted]----+-------------------------+--------------------+
| patient_id | first_name | last_name  | date_of_birth | gender | address                 | phone_number       |
+------------+------------+------------+---------------+--------+-------------------------+--------------------+
|          1 | John       | Doe        | 1980-05-15    | Male   | 123 Main St, Helsinki   | +358 40 1234567    |
|          2 | Jane       | Smith      | 1990-08-22    | Female | 456 Oak Ave, Espoo      | +358 50 9876543    |
+------------+------------+------------+---------[redacted]+-------------------------+--------------------+
"C:\UniServerZ\core\mysql\bin\mysql.exe" -u root [redacted] hospital -e "SELECT * FROM Staff;"
+----------+---------------+-----------+---------[redacted]----+-----------------------------+
| staff_id | first_name    | last_name | role         | specialty   | contact_info                |
+----------+---------------+-----------+--------------+-------------+-----------------------------+
|        1 | Dr. Michael   | Johnson   | Doctor       | Cardiology  | michael.johnson@hospital.fi |
|        2 | Dr. Sarah     | Williams  | Doctor       | Pediatrics  | sarah.williams@hospital.fi  |
|        3 | Nurse Mary    | Brown     | Nurse        | ICU         | mary.brown@hospital.fi      |
+----------+---------------+-----------+---------[redacted]+-------------+-----------------------------+
"C:\UniServerZ\core\mysql\bin\mysql.exe" -u root [redacted] hospital -e "SELECT * FROM Departments;"
+---------------+-----------------+---------[redacted]----+-----------------+
| department_id | department_name | location             | head_doctor_id  |
+---------------+-----------------+----------------------+-----------------+
|             1 | Cardiology      | Building A, Floor 3  |               1 |
|             2 | Pediatrics      | Building B, Floor 2  |               2 |
+---------------+-----------------+---------[redacted]----+-----------------+
"C:\UniServerZ\core\mysql\bin\mysql.exe" -u root [redacted] hospital -e "SELECT * FROM PatientStays;"
+---------+------------+---------[redacted]----+---------------+-------------------+
| stay_id | patient_id | department_id | admitting_doctor_id | admission_date    | discharge_date    |
+---------+------------+---------[redacted]----+---------------+-------------------+
|       1 |          1 |             1 |                   1 | 2026-03-01 09:00:00 | 2026-03-10 11:00:00 |
|       2 |          2 |             2 |                   2 | 2026-03-05 14:30:00 | NULL              |
+---------+------------+---------[redacted]----+---------------+-------------------+
"C:\UniServerZ\core\mysql\bin\mysql.exe" -u root [redacted] hospital -e "SELECT * FROM Treatments;"
+--------------+---------+---------[redacted]----+-------------------+-------------------------------+
| treatment_id | stay_id | staff_id | treatment_name   | treatment_date    | description                   |
+--------------+---------+---------[redacted]----+-------------------+-------------------------------+
|            1 |       1 |        1 | Heart Surgery    | 2026-03-02 10:00:00 | Successful heart bypass surgery |
|            2 |       2 |        2 | Vaccination      | 2026-03-05 15:00:00 | Routine childhood vaccination   |
+--------------+---------+---------[redacted]----+-------------------+-------------------------------+
"C:\UniServerZ\core\mysql\bin\mysql.exe" -u root [redacted] hospital -e "SELECT * FROM Medications;"
+---------------+-----------------+---------[redacted]----+-------------------------+
| medication_id | medication_name | dosage               | special_instructions    |
+---------------+-----------------+----------------------+-------------------------+
|             1 | Aspirin         | 100mg daily          | Take with food          |
|             2 | Antibiotics     | 500mg twice daily    | Complete full course    |
|             3 | Vitamin D       | 1000 IU daily        | Take in the morning     |
+---------------+-----------------+---------[redacted]----+-------------------------+
"C:\UniServerZ\core\mysql\bin\mysql.exe" -u root [redacted] hospital -e "SELECT * FROM Prescriptions;"
+------------------+------------+---------[redacted]----+---------[redacted]----+-------------------+-------------+-----------+
| prescription_id  | patient_id | medication_id | prescribing_doctor_id | prescription_date | start_date  | end_date  |
+------------------+------------+---------[redacted]----+---------[redacted]----+-------------------+-------------+-----------+
|                1 |          1 |             1 |                     1 | 2026-03-02        | 2026-03-02  | 2026-04-02 |
|                2 |          2 |             2 |                     2 | 2026-03-05        | 2026-03-05  | 2026-03-12 |
+------------------+------------+---------[redacted]----+---------[redacted]----+-------------------+-------------+-----------+
"C:\UniServerZ\core\mysql\bin\mysql.exe" -u root [redacted] hospital -e "SELECT * FROM Appointments;"
+------------------+------------+---------[redacted]----+---------[redacted]----+---------------------------+-------------+
| appointment_id   | patient_id | doctor_id | appointment_datetime | reason                    | status      |
+------------------+------------+---------[redacted]----+---------[redacted]----+---------------------------+-------------+
|                1 |          1 |         1 | 2026-03-15 10:00:00  | Follow-up after surgery   | Scheduled   |
|                2 |          2 |         2 | 2026-03-12 14:00:00  | Regular checkup           | Scheduled   |
+------------------+------------+---------[redacted]----+---------[redacted]----+---------------------------+-------------+
```

1.
"C:\UniServerZ\core\mysql\bin\mysql.exe" -u root -p hospital -e "INSERT INTO Patients (first_name, last_name, date_of_birth, gender, address, phone_number) VALUES ('Test', 'Patient', '2026-03-01', 'InvalidGender', 'Test Address', '123456789');" ERROR 1265 (01000) at line 1: Data truncated for column 'gender' at row 1 Selitys: Järjestelmä estää virheellisen sukupuolen arvon syöttämisen, koska gender-sarakkeessa on vain sallitut arvot 'Male', 'Female' tai 'Other'.

2.
"C:\UniServerZ\core\mysql\bin\mysql.exe" -u root -p hospital -e "INSERT INTO PatientStays (patient_id, department_id, admitting_doctor_id, admission_date) VALUES (999, 1, 1, '2026-03-01 09:00:00');" ERROR 1452 (23000) at line 1: Cannot add or update a child row: a foreign key constraint fails (`hospital`.`patientstays`, CONSTRAINT

`patientstays_ibfk_1` FOREIGN KEY (`patient_id`) REFERENCES `patients` (`patient_id`)) Selitys: Ulkomaantunnelet eivät salli viittausta olemattomaan potilaaseen. Potilaan ID 999 ei ole olemassa, joten lisäys estetään.

3. "C:\UniServerZ\core\mysql\bin\mysql.exe" -u root -p hospital -e "INSERT INTO Appointments (patient_id, doctor_id, appointment_datetime, reason, status) VALUES (1, 1, '2026-03-15 10:00:00', 'Test appointment', 'InvalidStatus');" ERROR 1265 (01000) at line 1: Data truncated for column 'status' at row 1 Selitys: Tilanumerointi (status) on rajoitettu vain sallittuihin arvoihin 'Scheduled', 'Completed' tai 'Cancelled'. Muita arvoja ei voida syöttää.

4. "C:\UniServerZ\core\mysql\bin\mysql.exe" -u root -p hospital -e "INSERT INTO Patients (first_name, last_name, date_of_birth, gender, address, phone_number) VALUES ('Future', 'Baby', '2027-01-01', 'Male', 'Future Address', '123456789');" Query OK, 1 row affected (0.01 sec) Huomautus: Vaikka syntymäpäivä tulevaisuudessa ei ole loogista, tietokanta ei estä sitä ilman erityistä CHECK-rajoitetta. Tämä voitaisiin korjata lisäämällä tarkistus.